Oxford College of Emory University

Chapter 14, Part 1

Money, Liquidity, and Banking

Guilherme Paiva Pinto

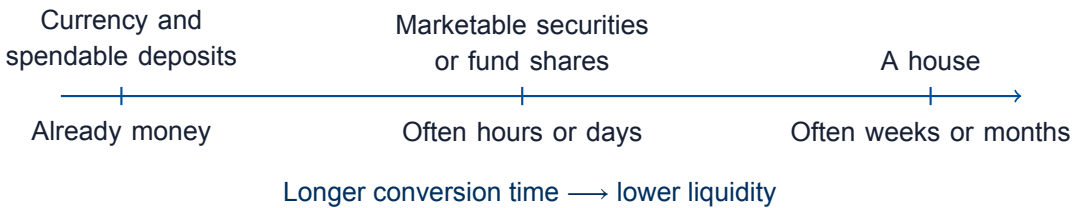
October 27, 2026

Roadmap

1. Money, liquidity, and the money supply
2. Banks, reserves, and T-accounts
3. Bank runs and safeguards
4. Moral hazard and shadow banking

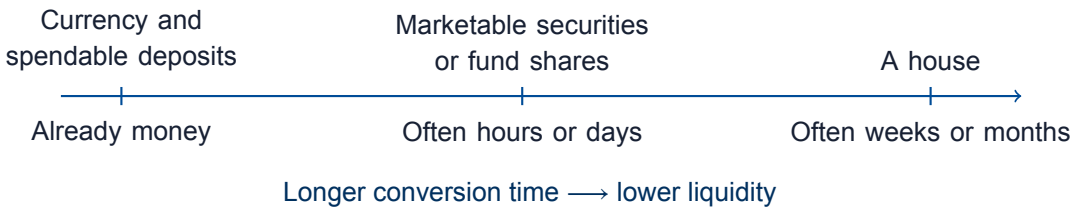
Money and Liquidity

- **Money**: what people readily accept as payment for goods and services.



Money and Liquidity

- **Money**: what people readily accept as payment for goods and services.
- **Liquidity**: how quickly something can become spendable money, with little cost or loss of value.



The Money Supply: M1 and M2

- **Money supply**: the total amount of currency and bank balances counted as money.

M2 = M1 + additional near-money. Near-money is readily convertible into spendable balances.

Source: Federal Reserve Board, H.6 definitions.

The Money Supply: M1 and M2

- **Money supply**: the total amount of currency and bank balances counted as money.
- **M1, the narrower measure**: public currency, demand deposits, and other liquid deposits, including savings deposits.

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The Money Supply: M1 and M2

- **Money supply**: the total amount of currency and bank balances counted as money.
- **M1, the narrower measure**: public currency, demand deposits, and other liquid deposits, including savings deposits.
- **M2, the broader measure**: M1 plus small time deposits and retail money market mutual fund balances.

M2 = M1 + additional near-money. Near-money is readily convertible into spendable balances.

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The Functions of Money

- **Medium of exchange**: an accepted way to pay for goods and services.
  - Pay for groceries without a double coincidence of wants.

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  - Pay for groceries without a double coincidence of wants.
- **Unit of account**: a common unit for stating and comparing values.
  - Compare a $5 smoothie with a $10 sandwich.
- **Store of value**: a way to carry purchasing power into the future.
  - Save today's earnings for next month's groceries. Inflation can erode their value.

Types of Money

A farmer brings produce to market. Compare three ways to pay.



Producing crops: Bruegel, *The Harvesters*, 1565



Trading: Van Ruysdael, *Market by the Seashore*, 1637

Commodity money

Pay with a portion of the crop accepted as money. It also has value as food.

Images: The Met, 19.164 and The Met, 60.55.4; both CC0.

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Commodity-backed money
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Fiat money
Pay with officially accepted money that promises no commodity redemption.

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What Banks Do

- Banks **accept deposits** and provide payment services.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

What Banks Do

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- They keep part of their funding in **reserves** and lend a large portion or buy investments such as bonds.

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What Banks Do

- Banks **accept deposits** and provide payment services.
- They keep part of their funding in **reserves** and lend a large portion or buy investments such as bonds.
- Deposits can be withdrawn quickly; many loans and investments repay over years.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Bank Reserves

- **Bank reserves**: cash in bank vaults plus the bank's deposits at the Federal Reserve.

Source: Federal Reserve Board, The Fed Explained.
Source: Federal Reserve Board, Reserve Requirements.

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- Reserves let a bank meet withdrawals and settle payments to other banks.

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Bank Reserves

- **Bank reserves**: cash in bank vaults plus the bank's deposits at the Federal Reserve.
- Reserves let a bank meet withdrawals and settle payments to other banks.
- **Reserve ratio**: bank reserves divided by the bank's deposits.

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T-Accounts

A **T-account** shows **what a bank has on the left** and **what it owes on the right**.

What the bank has		What the bank owes	
Loans	$1,200,000	Deposits	$1,000,000
Reserves	$100,000		

Total on the left: $1,300,000. Deposits: $1,000,000. The $300,000 difference is owners' money, which absorbs losses first.

Numerical example: Krugman and Wells, Macroeconomics, 7th ed., Chapter 14.

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Reserve ratio $= \frac{100{,}000}{1{,}000{,}000} = 10\%.$

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Money and Banking Checkpoint

A customer deposits $100 of currency into a checking account.

What changes for the bank and for public currency + deposits?

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Reserves +$100; deposits +$100.

Money and Banking Checkpoint

A customer deposits $100 of currency into a checking account.

What changes for the bank and for public currency + deposits?

Reserves +$100; deposits +$100.

Public currency −$100, so currency + deposits is unchanged.

Bank Runs

- A **bank run** happens when many customers rush to take out their money because they fear the bank may fail.

Source: Federal Reserve Board, SVB review, April 2023.

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- The bank has lent or invested much of that money. It cannot get it all back immediately.
- Selling investments quickly can mean losses. This can make more customers want to leave.

Source: Federal Reserve Board, SVB review, April 2023.

Silicon Valley Bank: March 2023

- **2022**: higher interest rates made SVB's older, long-term bonds worth less.

"Silicon Valley Bank shut by California regulator"
Reuters, March 10, 2023.

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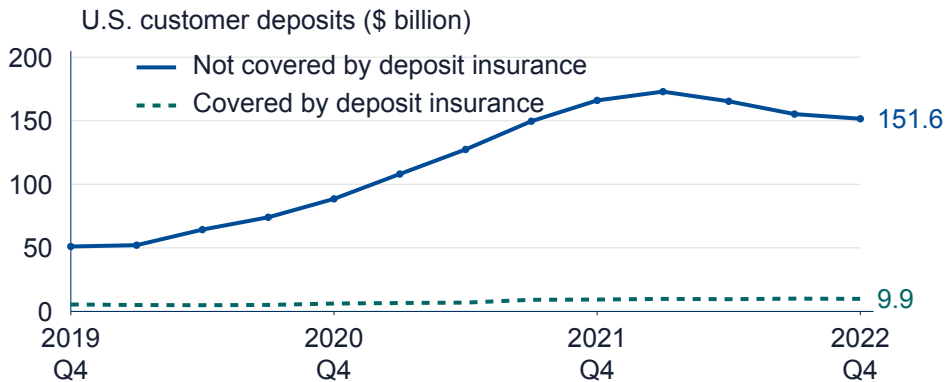
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- **March 8–9**: SVB announced losses. Worried customers withdrew more than $40 billion the next day.
- **March 10**: the bank could not meet the rush of withdrawals. Regulators closed it.

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Source: Federal Reserve Board, SVB review, April 2023.

Silicon Valley Bank: Customer Deposits



Original graph. Data: Federal Reserve Board, April 2023 SVB review, Figure 7 data; Call Report.

Bank Regulation

- **Deposit insurance**: the FDIC, created in 1933, protects eligible deposits up to $250,000 per depositor, insured bank, and ownership category.

Sources: FDIC history; deposit coverage; Federal Reserve: discount window, IORB, capital.

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- **Interest on reserve balances (IORB)**: the Fed pays interest on money banks keep in their accounts at the Fed.
- **Capital requirements**: owners must put in their own money to absorb losses first.

Sources: FDIC history; deposit coverage; Federal Reserve: discount window, IORB, capital.

Moral Hazard

- **Moral hazard**: protection from losses can encourage greater risk-taking.

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- Customers may watch risks less closely if their deposits are insured. Owners may gain from risky bets while others bear some losses.
- **Capital requirements** make owners risk more of their own money; bank oversight helps enforce the rules.

Source: Federal Reserve Board, regulatory capital.

Shadow Banking

- **Shadow banking**: lending and borrowing through firms that are not ordinary deposit banks.

Source: Financial Stability Board, Non-Bank Financial Intermediation.

Source: Federal Reserve Board, crisis institutions and funding runs (2009).

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- Examples: **investment banks** help companies raise money; **money market funds** pool savings for short-term lending.

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The 2008 crisis

- September 15: **Lehman Brothers** went bankrupt.
- September 16: losses at **Reserve Primary Fund** led investors to demand their money back.

Source: Financial Stability Board, Non-Bank Financial Intermediation.

Source: Federal Reserve Board, crisis institutions and funding runs (2009).

Chapter 14, Part 2

Oxford College of Emory University

Money Creation and Central Banks

Guilherme Paiva Pinto

October 29, 2026

Roadmap

1. Explain how banks create money.
2. Calculate and interpret the simple money multiplier.
3. Understand how central banks work and identify their policy tools.
4. Understand how lessons from financial crises shaped monetary policymaking.

Chapter 14, Part 2 · Section 1

Explain how banks create money.

Trace deposits, loans and payments, distinguishing changes in the money supply from changes in the monetary base.

Determining the Money Supply

Money supply = public currency + checkable deposits

Simplified measure: M = C + D

The same $1,000	Currency (C)	Deposits (D)	Money (M)
Before depositing	$1,000	$0	$1,000
After depositing			

Determining the Money Supply

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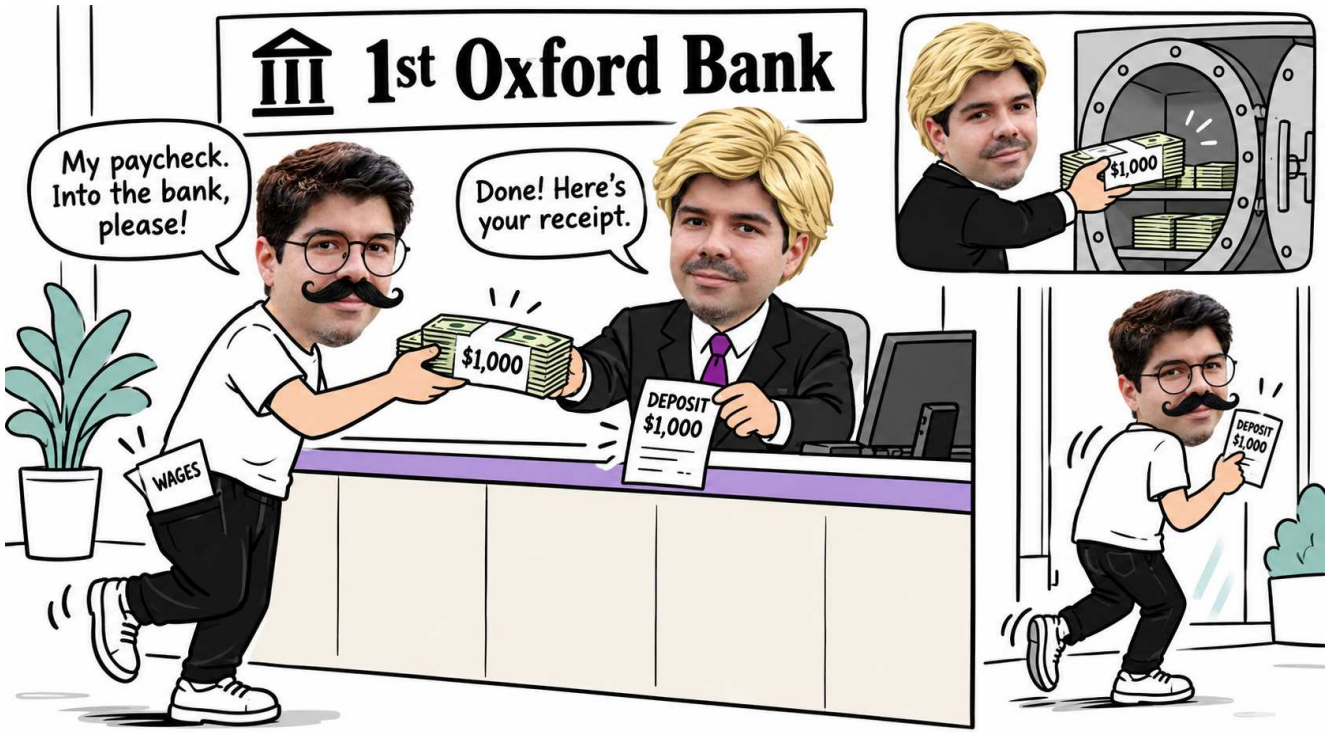
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The same $1,000	Currency (C)	Deposits (D)	Money (M)
Before depositing	$1,000	$0	$1,000
After depositing	$0	$1,000	$1,000

The deposit changes the form of money, not its total.

How Banks Create Money

Initial cash: $1,000 · Reserve ratio: rr = 10%



The cash deposit changes the form of $1,000, not the money total.

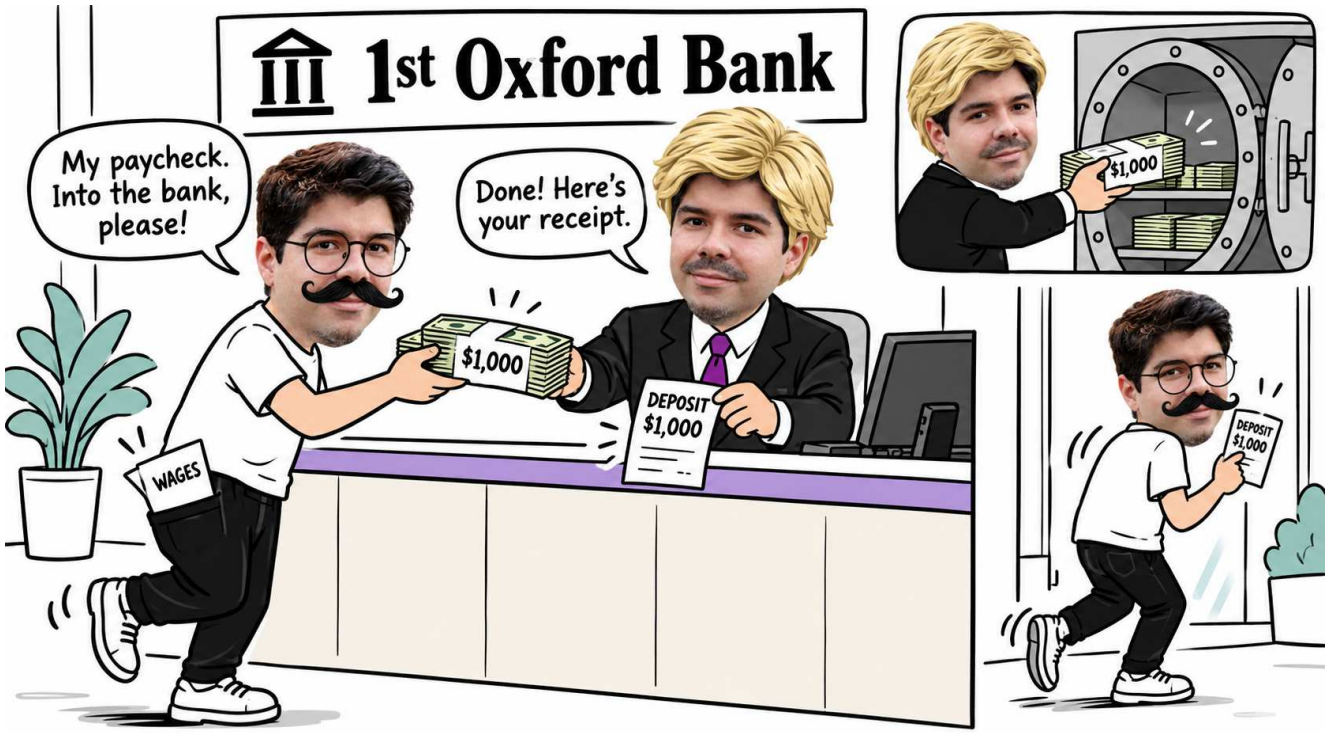
How banks create money

Stage	Currency in circulation	Checkable bank deposits	Money supply
First stage Cash held by the public	$1,000	$0	$1,000
Second stage First deposit → $900 loan and payment			
Third stage $900 redeposit → $810 new loan			

Numerical stages: Krugman & Wells, 7th ed., Chapter 14, Table 1. Each row includes the completed loan stage.

How Banks Create Money

Initial cash: $1,000 · Reserve ratio: rr = 10%



First bank: $100 required reserves; $900 available to lend.

How banks create money

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How Banks Create Money

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The first bank lends $900 in cash.

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Initial cash: $1,000 · Reserve ratio: rr = 10%



The borrower pays the same $900 to the supplier.

How banks create money

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Third stage $900 redeposit → $810 new loan	$810	$1,900	$2,710

The next bank receives $900, keeps $90 and lends $810.

The third stage includes the **next $810 loan**.

Numerical stages: Krugman & Wells, 7th ed., Chapter 14, Table 1. Each row includes the completed loan stage.

Chapter 14, Part 2 · Section 2

Calculate and interpret the simple money multiplier.

Determine total deposit expansion and explain the assumptions that limit the model’s application to actual banking.

Reserves and Repeated Deposit Creation

Excess reserves = reserves − required reserves.

rr = 10% · Each row follows lending and payment to the next bank.

Bank	Deposits	Reserves kept	New loan
A	$1,000		
B			
C			
…	…	…	…

Reserves and Repeated Deposit Creation

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At Bank A: $1,000 − (0.10 × $1,000) = **$900**.

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A	$1,000	$100	$900
B	$900	$90	$810
C	$810	$81	$729
…	…	…	…

The Simple Money Multiplier

Initial cash deposit: \$1,000 · rr = 0.10

$$D = 1{,}000 + 1{,}000(1 - rr) + 1{,}000(1 - rr)^2 + \ldots$$

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$10,000 total deposits includes the original $1,000.

$9,000 additional deposits are created through lending.

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Under these assumptions, \$1 of reserves supports \$10 of deposits.

Why Reality Is More Complicated

Full redeposit. People deposit the payments they receive.

Bank of England, Money Creation (2014) · Federal Reserve, Reserve Requirements

Why Reality Is More Complicated

Full redeposit. People deposit the payments they receive.

Only required reserves are retained. Banks keep rr × deposits and lend the remaining amount.

At rr = 10%: a $900 deposit → $90 reserves + $810 loan.

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Why Reality Is More Complicated

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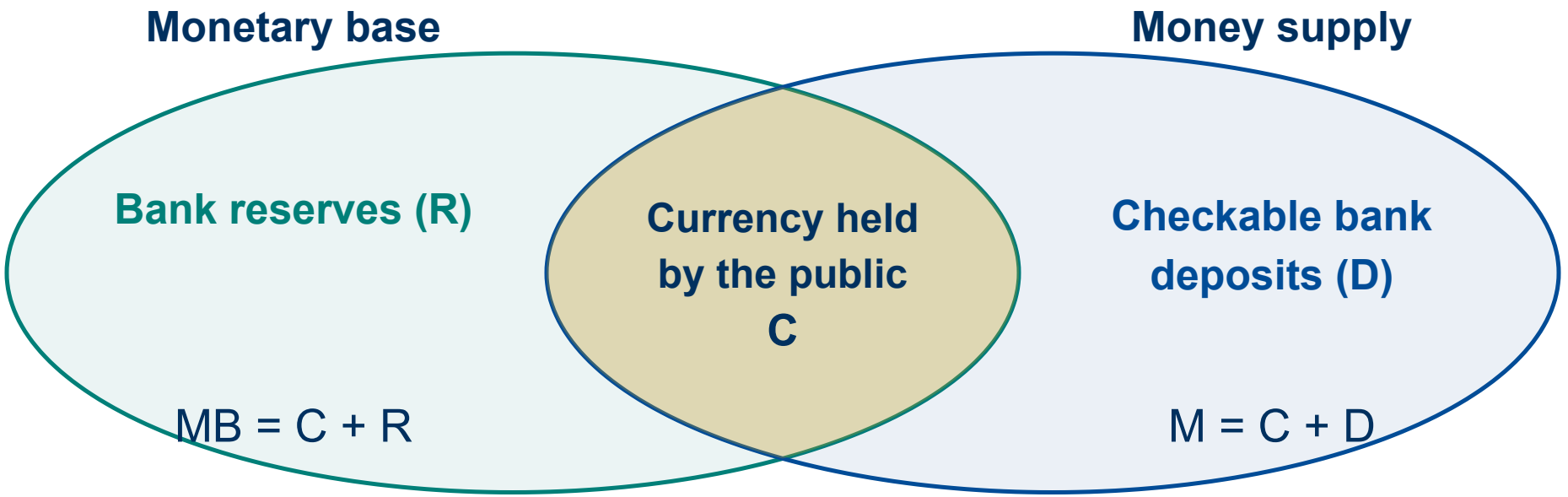
More cash held by the public or more excess reserves
means less deposit expansion.

1/rr is the model's simple deposit multiplier.

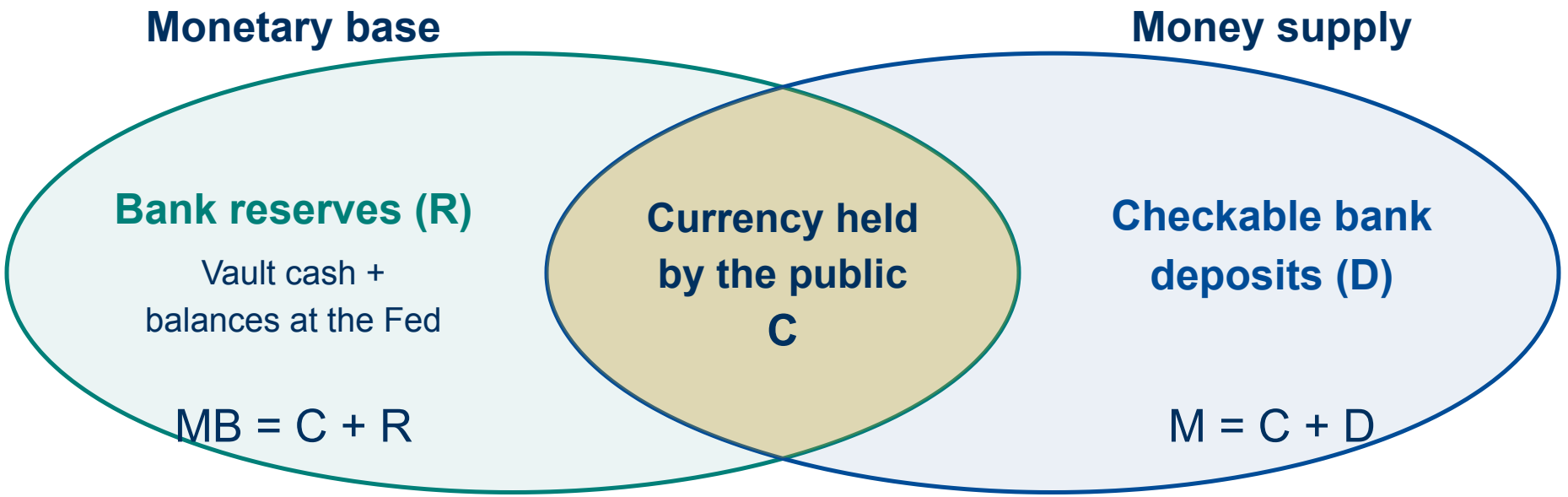
rr = 10% is our example; the U.S. statutory requirement is 0%.

Bank of England, Money Creation (2014) · Federal Reserve, Reserve Requirements

Monetary Base and Money Supply

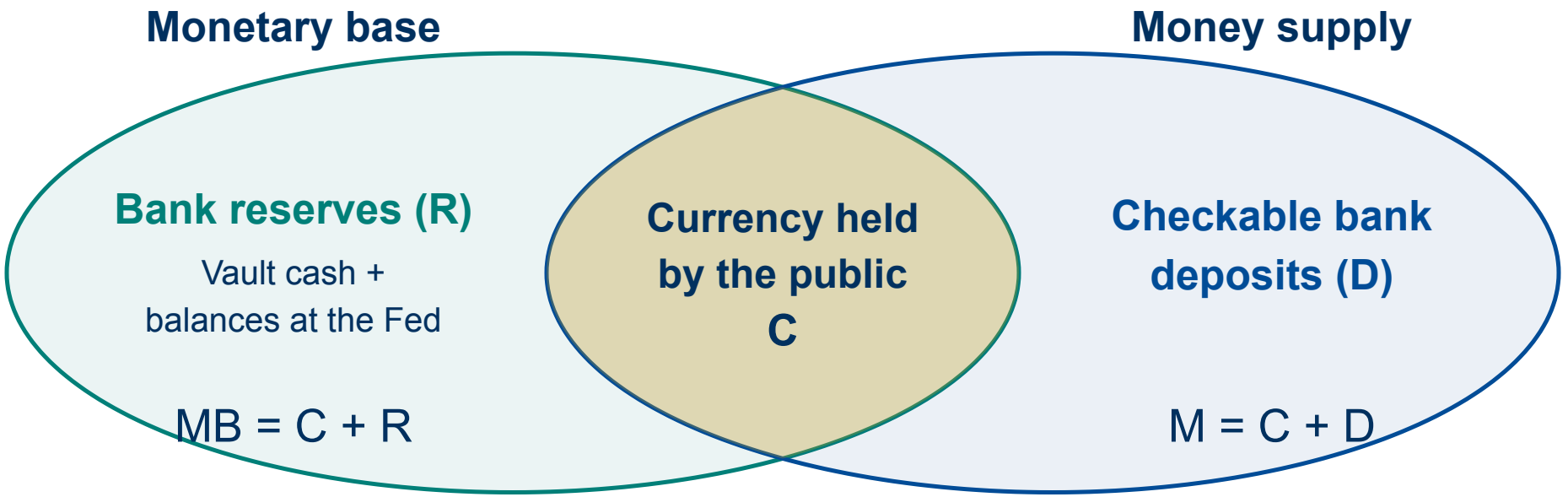


Monetary Base and Money Supply



Public currency is the overlap. Bank reserves and public deposits are different components.

Monetary Base and Money Supply



Public currency is the overlap. Bank reserves and public deposits are different components.

$$\text{Money multiplier} = \frac{M}{MB}$$

Next: how Fed operations change reserves and the monetary base.

Money Creation Checkpoint

$1,000 of existing public cash · rr = 10% · Full lending and redeposit

1. Cash becomes a deposit.

Does the money supply rise immediately?

2. Lending repeats across banks.

What are total deposits and the net increase in money?

3. Banks make loans.

Does lending itself create monetary base?

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Deposits: $10,000. Net increase in M: $9,000.

3. Banks make loans.

Does lending itself create monetary base?

No. It creates deposits; it does not create base money.

Chapter 14, Part 2 · Section 3

Understand how central banks work and identify their policy tools.

Explain the roles of the discount rate, interest on reserve balances and open-market operations.

The Federal Reserve System



1. Boston
2. New York
3. Philadelphia
4. Cleveland
5. Richmond
6. Atlanta
7. Chicago
8. St. Louis
9. Minneapolis
10. Kansas City
11. Dallas
12. San Francisco

Reserve Bank locations. San Francisco also serves Alaska and Hawaii.

Board of Governors

7 seats · Washington, D.C.

Nominated by the President; confirmed by the Senate.

Federal Reserve, System Structure · Map: Natural Earth

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Board of Governors

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Nominated by the President; confirmed by the Senate.

Oversees the System. Governors serve staggered 14-year terms.

Federal Reserve, System Structure · Map: Natural Earth

The Federal Open Market Committee

FOMC = the Fed’s monetary-policy committee.

7	+	1	+	4
Board governor seats		New York Fed president		Rotating Reserve Bank presidents

Federal Reserve, FOMC · Federal Reserve, System Structure

The Federal Open Market Committee

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Board governor seats		New York Fed president		Rotating Reserve Bank presidents

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All Reserve Bank presidents participate; voting rights rotate.
The FOMC does not set every household or business loan rate.

Federal Reserve, FOMC · Federal Reserve, System Structure

What the Fed Does

Monetary policy · financial stability and supervision · payments

The federal funds market

Overnight, unsecured lending of balances at the Fed between banks and other eligible institutions.

New York Fed, Monetary Policy Implementation

What the Fed Does

Monetary policy · financial stability and supervision · payments

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FFR: the federal funds rate—the interest rate in this market.

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What the Fed Does

Monetary policy · financial stability and supervision · payments

The federal funds market

Overnight, unsecured lending of balances at the Fed between banks and other eligible institutions.

FFR: the federal funds rate—the interest rate in this market.

Policy tool	What changes
Discount rate	Cost of borrowing from the Fed
Interest on reserve balances	Return on eligible balances at the Fed
Open-market operations	Securities and bank reserve balances

Next week: interest rates, spending, and aggregate demand.

New York Fed, Monetary Policy Implementation

The Discount Window and Reserve Interest

Discount window

The Fed lends to eligible banks against collateral.

Interest on reserve balances

IORB pays banks on eligible balances held at the Fed.

Federal Reserve, Discount Window · Federal Reserve, IORB FAQ

The Discount Window and Reserve Interest

Discount window

The Fed lends to eligible banks against collateral.

Lower discount rate:
Cheaper Fed funding; borrowing becomes more attractive.

Higher discount rate:
Costlier Fed funding; borrowing becomes less attractive.

Interest on reserve balances

IORB pays banks on eligible balances held at the Fed.

Federal Reserve, Discount Window · Federal Reserve, IORB FAQ

The Discount Window and Reserve Interest

Discount window

The Fed lends to eligible banks against collateral.

Lower discount rate:
Cheaper Fed funding; borrowing becomes more attractive.

Higher discount rate:
Costlier Fed funding; borrowing becomes less attractive.

Interest on reserve balances

IORB pays banks on eligible balances held at the Fed.

A reservation rate:
a safe alternative return to lending.

Higher IORB raises that benchmark; lower IORB reduces it.

IORB influences lending rates; it is not a fixed minimum rate on every loan.

Federal Reserve, Discount Window · Federal Reserve, IORB FAQ

Open-Market Purchases

The Fed buys a $1,000 Treasury security from a commercial bank.



Purchase: reserves enter the bank.

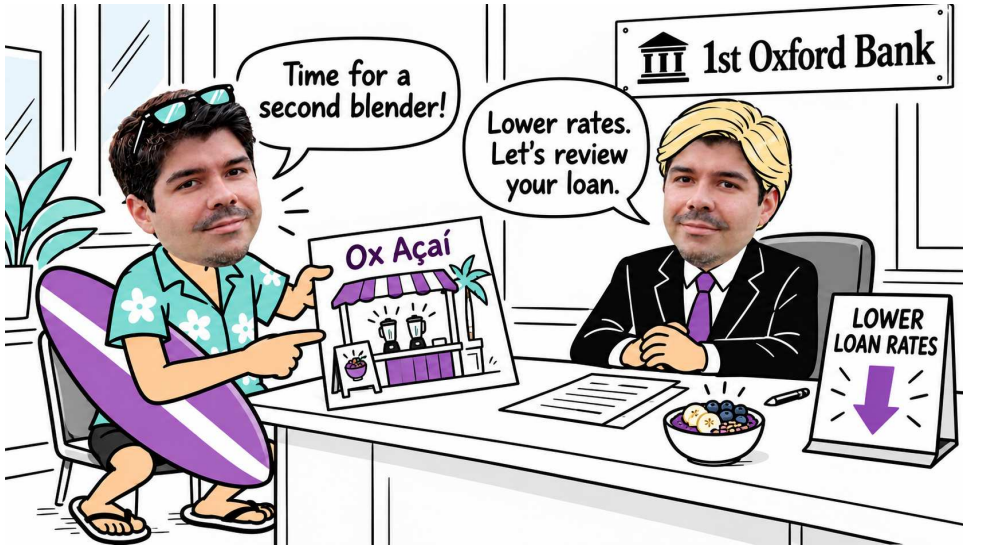
New York Fed, Monetary Policy Implementation

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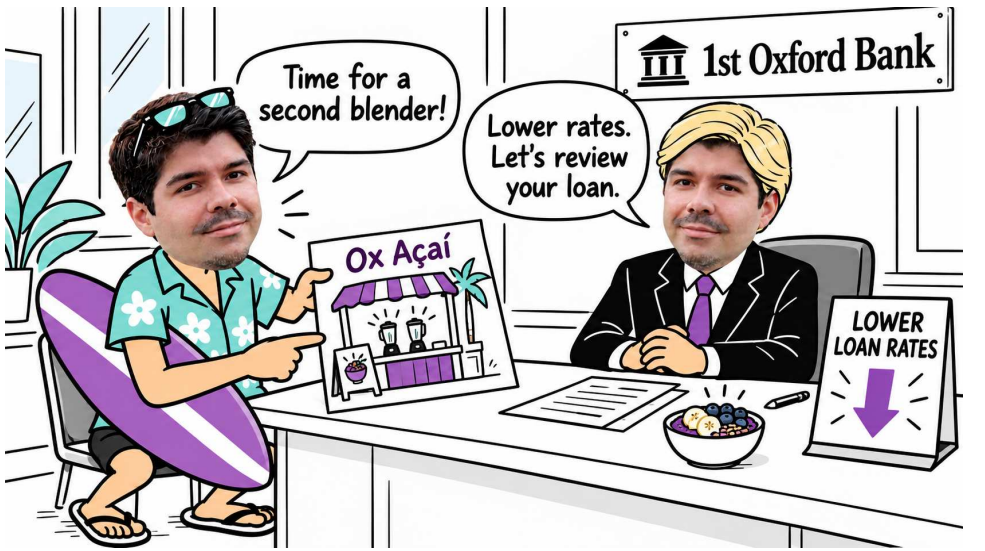
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Bank reserves +$1,000 · Monetary base +$1,000

Bank securities −$1,000 · Public deposits: no immediate change.

Loan-rate effects are conditional; reserve changes alone do not guarantee them.

New York Fed, Monetary Policy Implementation

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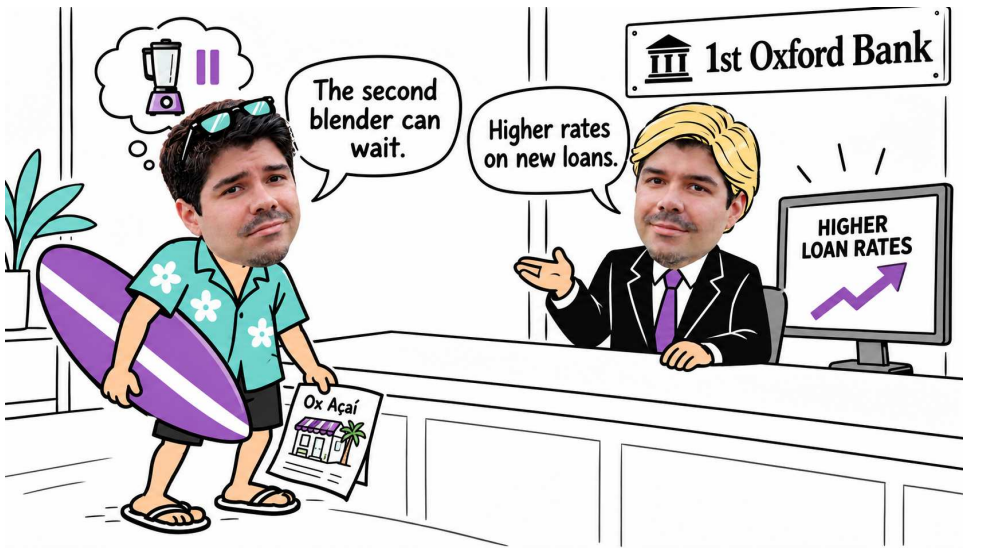
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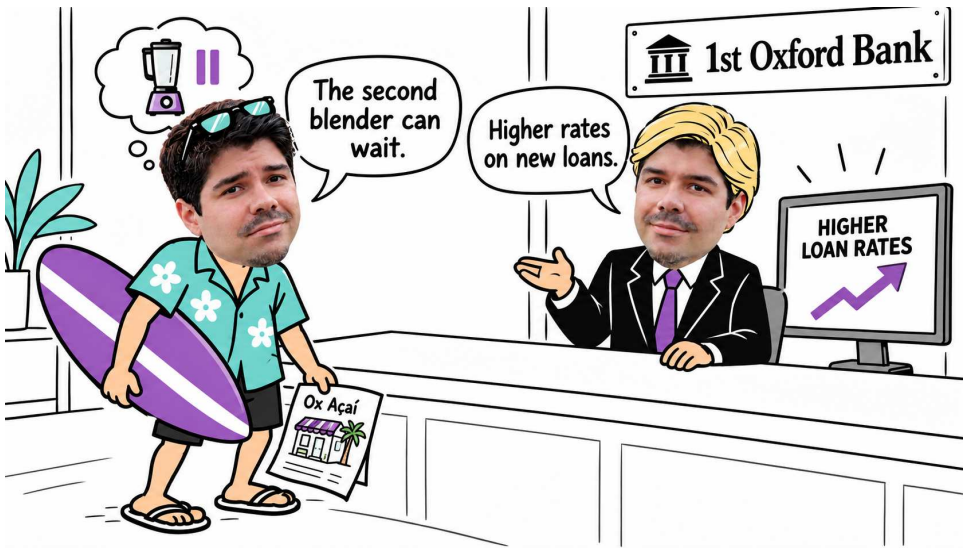
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New York Fed, Monetary Policy Implementation

The ECB and the Federal Reserve

ECB = European Central Bank · Eurosystem = ECB + euro-area national central banks.

	Federal Reserve	ECB / Eurosystem
Central board	7 Board governor seats	6 Executive Board members
Policy committee		
Operating network		
Main policy goals		

ECB, Governing Council · ECB, Eurosystem · ECB, Monetary Policy · ECB, Standing Facilities · Federal Reserve, System Structure

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Main policy goals	Maximum employment; stable prices	Price stability is primary

Both use policy rates, lending facilities, and asset operations.

The ECB's deposit facility offers a comparable remuneration benchmark.

ECB, Governing Council · ECB, Eurosystem · ECB, Monetary Policy · ECB, Standing Facilities · Federal Reserve, System Structure

Chapter 14, Part 2 · Section 4

Understand how lessons from financial crises shaped monetary policymaking.

Connect historical crises to changes in central banks’ responsibilities and policy approaches.

Banking Panics and the Creation of the Fed

1907: banking panic

Withdrawal demands outran available cash; emergency support depended on private coordination.

Fed History: Panic of 1907 · Fed History: Federal Reserve Act · Fed History: Great Depression

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A central reserve system could supply liquidity when individual banks could not.

1930–1933: renewed bank runs

The Fed did not prevent a severe monetary contraction. The 1933 Banking Act created federal deposit insurance.

Policy lesson: Central-bank liquidity and deposit protection address different sources of banking instability.

Fed History: Panic of 1907 · Fed History: Federal Reserve Act · Fed History: Great Depression

The Savings-and-Loan Crisis

1980s: a funding mismatch

Thrifts held long-term, fixed-rate mortgages while rising rates increased their funding costs.

Fed History: Savings-and-Loan Crisis · FDIC: 1980–1989

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Risk-taking and weak oversight

Broader investment powers, weak capital and insured deposits encouraged costly risk-taking.

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Broader investment powers, weak capital and insured deposits encouraged costly risk-taking.

1989: cleanup and reform

FIRREA reorganized thrift oversight and resolution. Public funds helped cover the failed institutions' losses.

Policy lesson: Deposit insurance needs capital, supervision and risk controls to limit excessive risk-taking.

Fed History: Savings-and-Loan Crisis · FDIC: 1980–1989

The 2008 Crisis and Its Aftermath

2007–2009: housing losses and funding runs

Mortgage-related losses and withdrawals of short-term funding spread stress beyond deposit banks.

Fed History: Great Recession · Federal Reserve: Dodd–Frank

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The Fed expanded liquidity lending and asset purchases; the Treasury provided capital support.

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2010: Dodd–Frank reforms

New systemic-risk oversight and consumer protection addressed gaps exposed by the crisis.

Policy lesson: Central-bank tools must address market-wide funding stress, beyond runs at deposit banks.

Fed History: Great Recession · Federal Reserve: Dodd–Frank

Reference Definitions and Examples

Money, income, wealth, and bank accounting.

Money, Income, and Net Wealth

Concept	Meaning	Example
Money	Readily spendable assets	$400 in currency and deposits
Income	Receipts over a period	$900 earned this month
Net wealth	Assets minus liabilities	$5,000 − $2,000 = $3,000

A Smoothie Company Balance Sheet

Assets	Liabilities and equity
Cash and inventory	Loans and other obligations
Equipment; a building if owned	Equity: the owners' residual claim

Assets = liabilities + equity.

Liquidity and Solvency

Issue	Question	Possible problem
Liquidity	Can the bank pay now?	Assets cannot all be sold quickly at little cost.
Solvency	Do assets cover liabilities?	Losses can exhaust equity.

Forced sales can turn liquidity pressure into losses.

A Cash Deposit

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

A customer deposits $200 of currency held outside banks.

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Reserves and deposits each rise by $200.

Public currency falls by $200; currency + deposits is unchanged.

A New Bank Loan

- The bank approves a $120 loan and credits the borrower's deposit.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

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A New Bank Loan

- The bank approves a $120 loan and credits the borrower's deposit.
- Loans and deposits each rise by $120 at origination.
- The borrower gains a deposit and owes a debt; net wealth does not rise by $120.

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Reserves and the Monetary Base

- **Public currency**: notes and coins outside banks.

Source: Federal Reserve Board, H.6 definitions.

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- **Public currency**: notes and coins outside banks.
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Reserves and the Monetary Base

- **Public currency**: notes and coins outside banks.
- **Bank reserves**: vault cash and balances at the Fed.
- **Monetary base**: public currency plus bank reserves.

Source: Federal Reserve Board, H.6 definitions.